SYSTEM DESIGN DOCUMENT

Cyber-Physical Smart Building Access Control System

Phase 1 Deliverable — Architecture, Firmware, Database, API, and UI Design

1. System Block Diagram

The System is organized into three layers: the edge (door node hardware), the network (hybrid RS-485 backbone with Wi-Fi fallback), and the application layer (backend, database, dashboard). This mirrors the layered architecture established in the revised proposal.

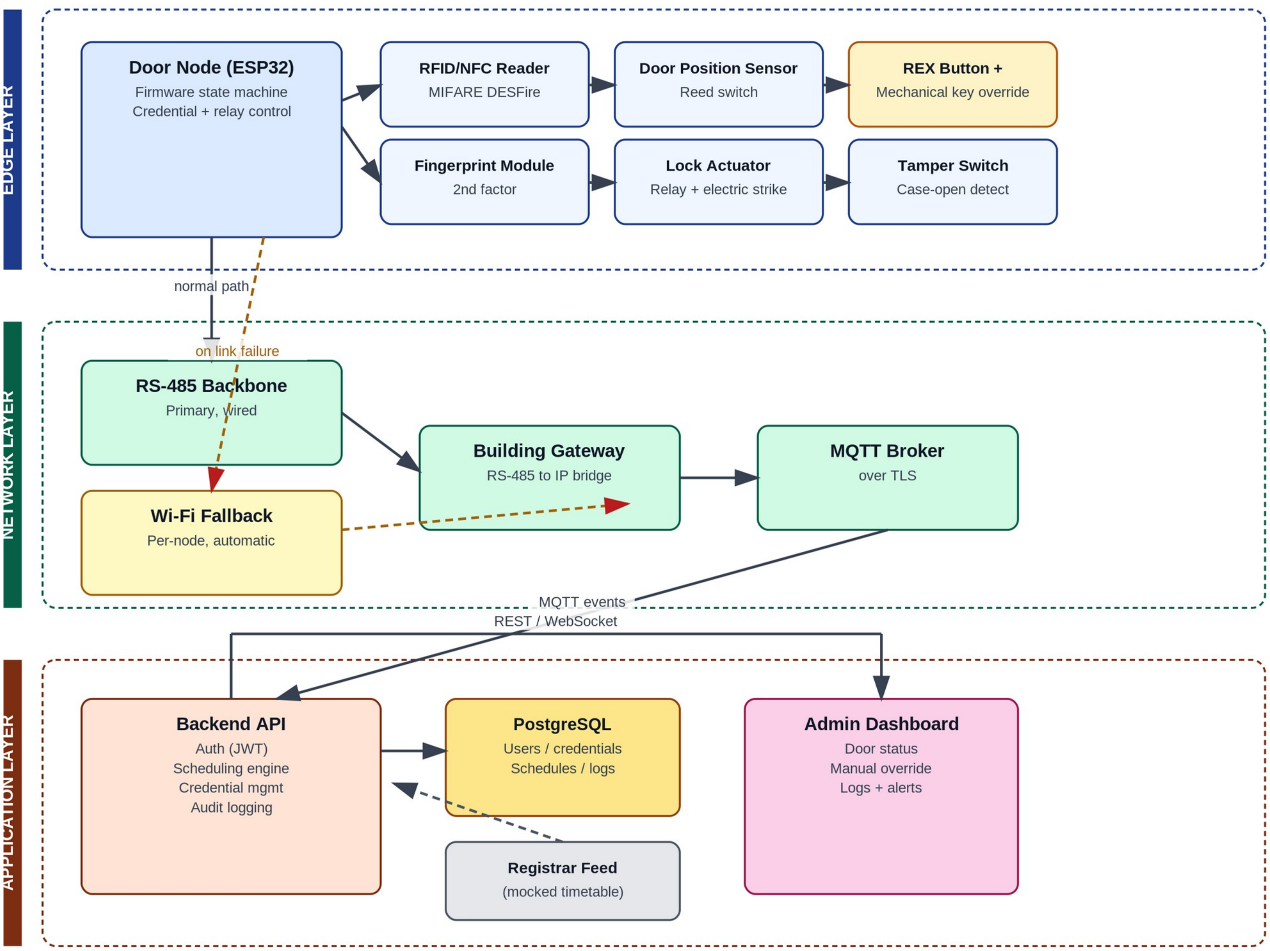

Figure 1 — Three-layer system architecture and data flow.

2. Door Node Wiring & Connection Diagram

Each door node centers on an ESP32 driving all peripherals through the interfaces below. This is a functional wiring diagram, not a fabrication-ready schematic — resistor values, connector pinouts, and PCB layout are produced in the next design pass once this topology is confirmed.

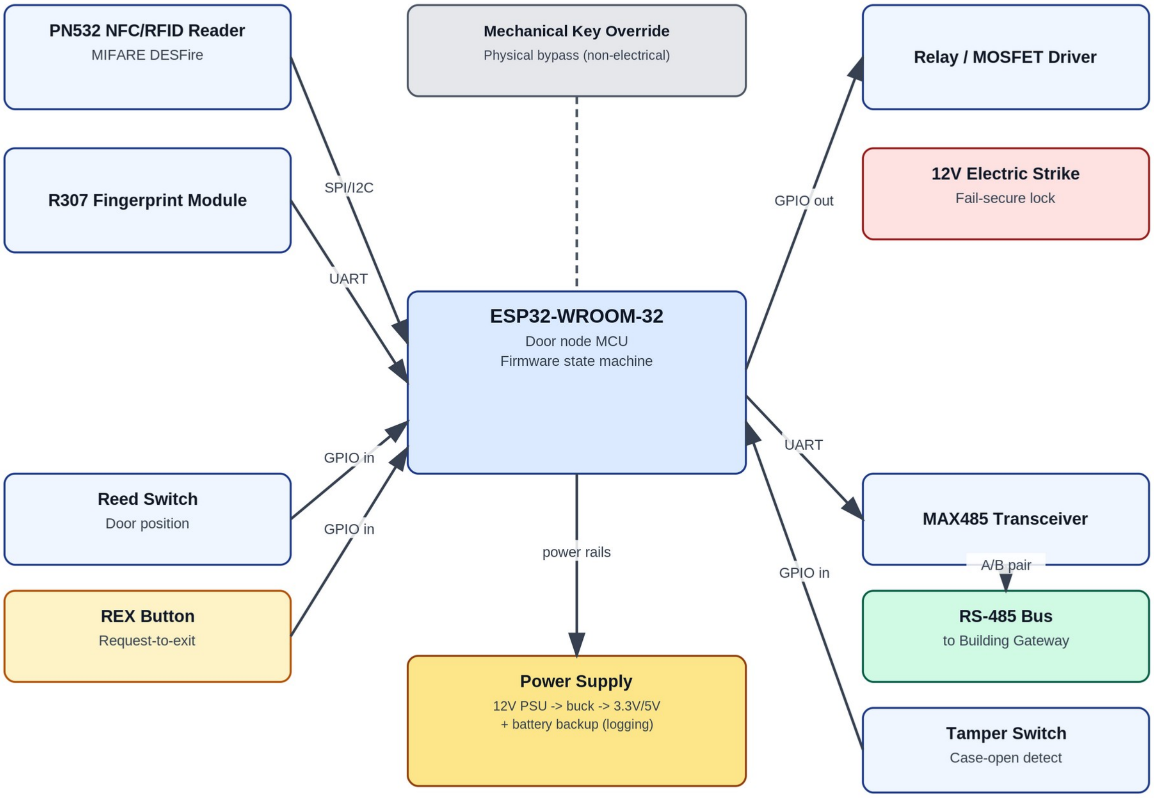

Figure 2 — ESP32 peripheral connections and protocols.

- SPI/I2C: PN532 NFC/RFID reader (configurable per hardware revision).
- UART: R307 fingerprint module and MAX485 RS-485 transceiver (separate UART peripherals on the ESP32).
- GPIO in: reed switch (door position), REX button, tamper switch.
- GPIO out: relay/MOSFET driver firing the electric strike.
- Mechanical key override is a physical bypass with no electrical tie to the ESP32 — it must work even if the node is fully powered down.

3. Firmware State Machine

The door node firmware runs a primary access state machine plus two independent background tasks: a network watchdog (handles RS-485/Wi-Fi failover) and a tamper interrupt that can fire from any state.

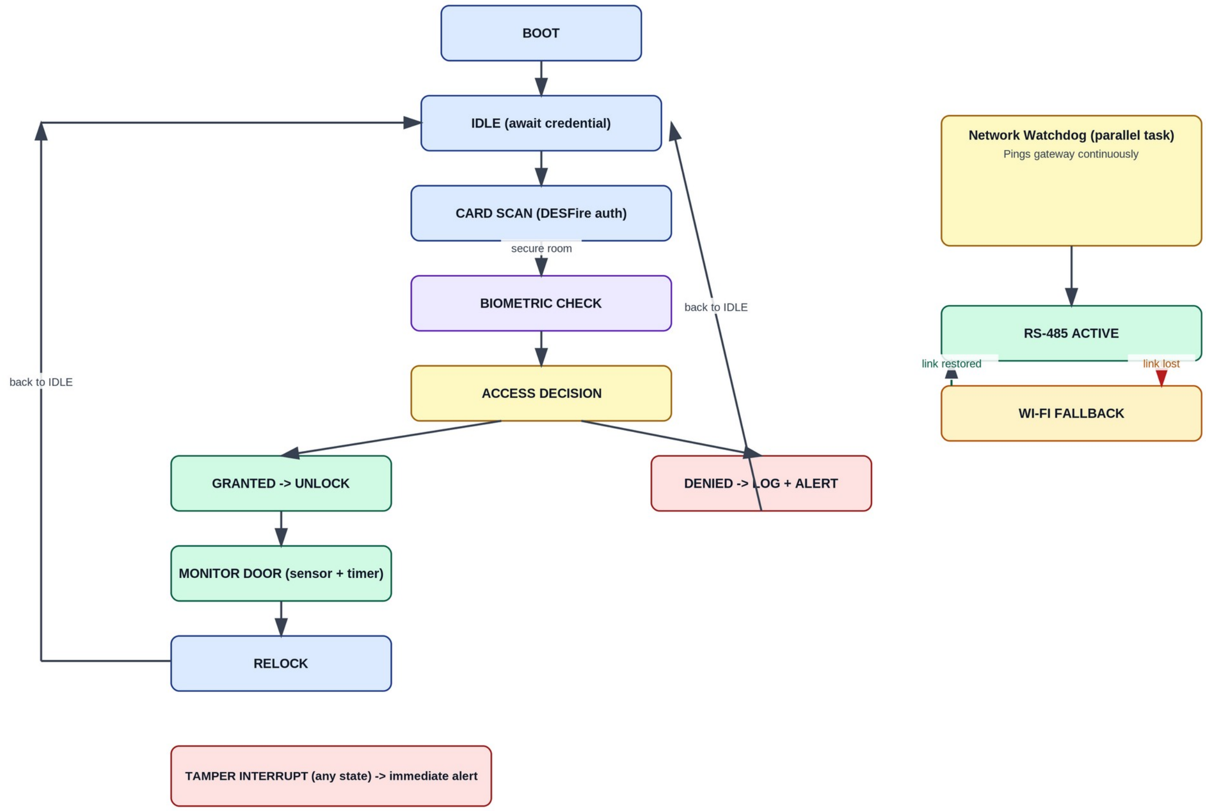

Figure 3 — Door node firmware states and transitions.

- IDLE waits for a card or fingerprint read; BIOMETRIC CHECK only runs for doors flagged as higher-security.
- MONITOR DOOR compares the commanded lock state against the reed switch reading, so a propped-open or forced door is detected independently of the lock's own status.
- The network watchdog runs continuously in parallel; a lost RS-485 link switches the node to its Wi-Fi radio without interrupting the access flow.

4. Database Schema

Six core tables cover users, credentials, doors, schedules, access events, and alerts. Foreign keys tie access events and alerts back to both the door and the credential involved, which is what the dashboard's history and audit views query against.

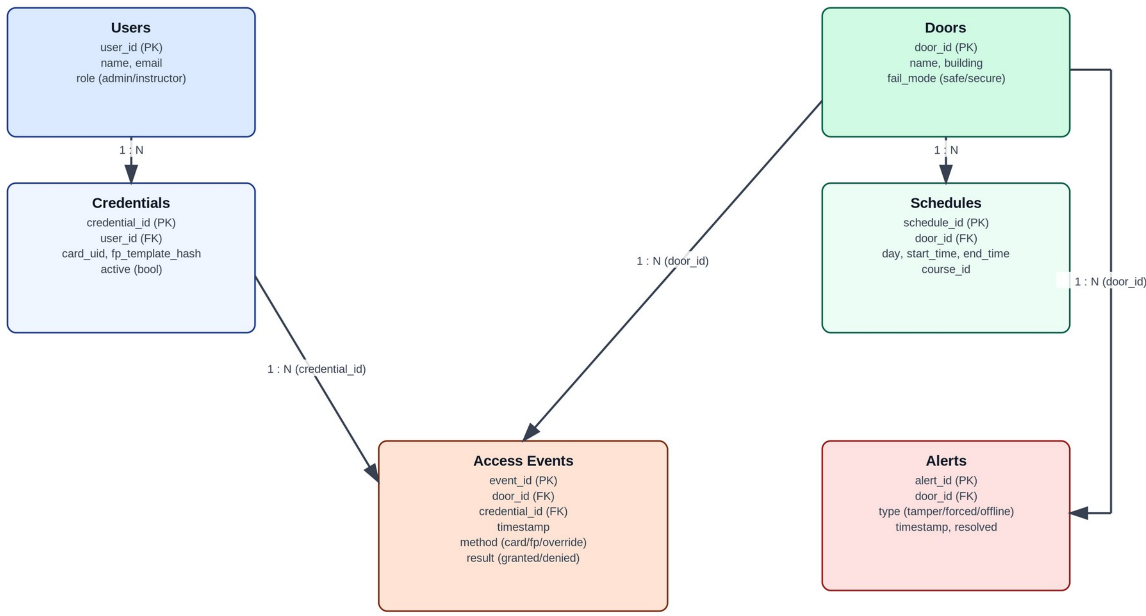

Figure 4 — Entity-relationship diagram.

```
CREATE TABLE users (
  user_id       SERIAL PRIMARY KEY,
  name          VARCHAR(120) NOT NULL,
  email         VARCHAR(160) UNIQUE NOT NULL,
  role          VARCHAR(20)  NOT NULL CHECK (role IN ('admin','instructor')),
  created_at    TIMESTAMP NOT NULL DEFAULT now()
);

CREATE TABLE credentials (
  credential_id     SERIAL PRIMARY KEY,
  user_id           INTEGER REFERENCES users(user_id),
  card_uid          VARCHAR(64),
  fp_template_hash  VARCHAR(256),
  active            BOOLEAN NOT NULL DEFAULT true,
  issued_at         TIMESTAMP NOT NULL DEFAULT now()
);

CREATE TABLE doors (
  door_id       SERIAL PRIMARY KEY,
  name          VARCHAR(80) NOT NULL,
  building      VARCHAR(80) NOT NULL,
  fail_mode     VARCHAR(10) NOT NULL CHECK (fail_mode IN ('safe','secure'))
);

CREATE TABLE schedules (
  schedule_id   SERIAL PRIMARY KEY,
  door_id       INTEGER REFERENCES doors(door_id),
  day_of_week   SMALLINT NOT NULL,
  start_time    TIME NOT NULL,
  end_time      TIME NOT NULL,
```

```
  course_id     VARCHAR(40)
);

CREATE TABLE access_events (
  event_id       BIGSERIAL PRIMARY KEY,
  door_id        INTEGER REFERENCES doors(door_id),
  credential_id  INTEGER REFERENCES credentials(credential_id),
  event_time     TIMESTAMP NOT NULL DEFAULT now(),
  method         VARCHAR(20) NOT NULL,
  result         VARCHAR(20) NOT NULL
);

CREATE TABLE alerts (
  alert_id      SERIAL PRIMARY KEY,
  door_id       INTEGER REFERENCES doors(door_id),
  type          VARCHAR(20) NOT NULL,
  alert_time    TIMESTAMP NOT NULL DEFAULT now(),
  resolved      BOOLEAN NOT NULL DEFAULT false
);
```

5. REST API Specification

Method	Endpoint	Description
POST	/api/auth/login	Authenticate admin/instructor; returns a JWT
POST	/api/auth/refresh	Refresh an expiring JWT
GET	/api/doors	List all doors with current status
GET	/api/doors/{door_id}	Get detail for a single door
POST	/api/doors/{door_id}/override	Manual unlock/lock override
GET	/api/doors/{door_id}/logs	Access event history for a door
GET	/api/schedules	List schedules
POST	/api/schedules	Create or import a schedule entry
PUT	/api/schedules/{schedule_id}	Update a schedule entry
GET	/api/credentials	List credentials (admin only)
POST	/api/credentials	Issue a new credential
DELETE	/api/credentials/{credential_id}	Revoke a credential
GET	/api/alerts	List active/resolved alerts
PUT	/api/alerts/{alert_id}/resolve	Mark an alert resolved
GET	/api/users	List users (admin only)

6. MQTT Topic Structure

Topic	Direction	Payload
site/{door_id}/status	Door → Backend	Retained online/offline heartbeat
site/{door_id}/event	Door → Backend	JSON access event (credential, method, result, timestamp)
site/{door_id}/alert	Door → Backend	Tamper / forced-entry / sensor-mismatch alert

Topic	Direction	Payload
site/{door_id}/cmd	Backend → Door	Override command (lock / unlock)
site/{door_id}/cmd/ack	Door → Backend	Command execution acknowledgment
site/gateway/health	Gateway → Backend	Building gateway heartbeat

All topics are published over MQTT with TLS enabled on the broker connection; door nodes authenticate to the broker with per-device credentials rather than a shared password.

7. Dashboard Wireframe

Low-fidelity layout for the admin dashboard's main screen: door status grid with per-door override/history actions, an alert banner for anything needing immediate attention, and a recent access events table.

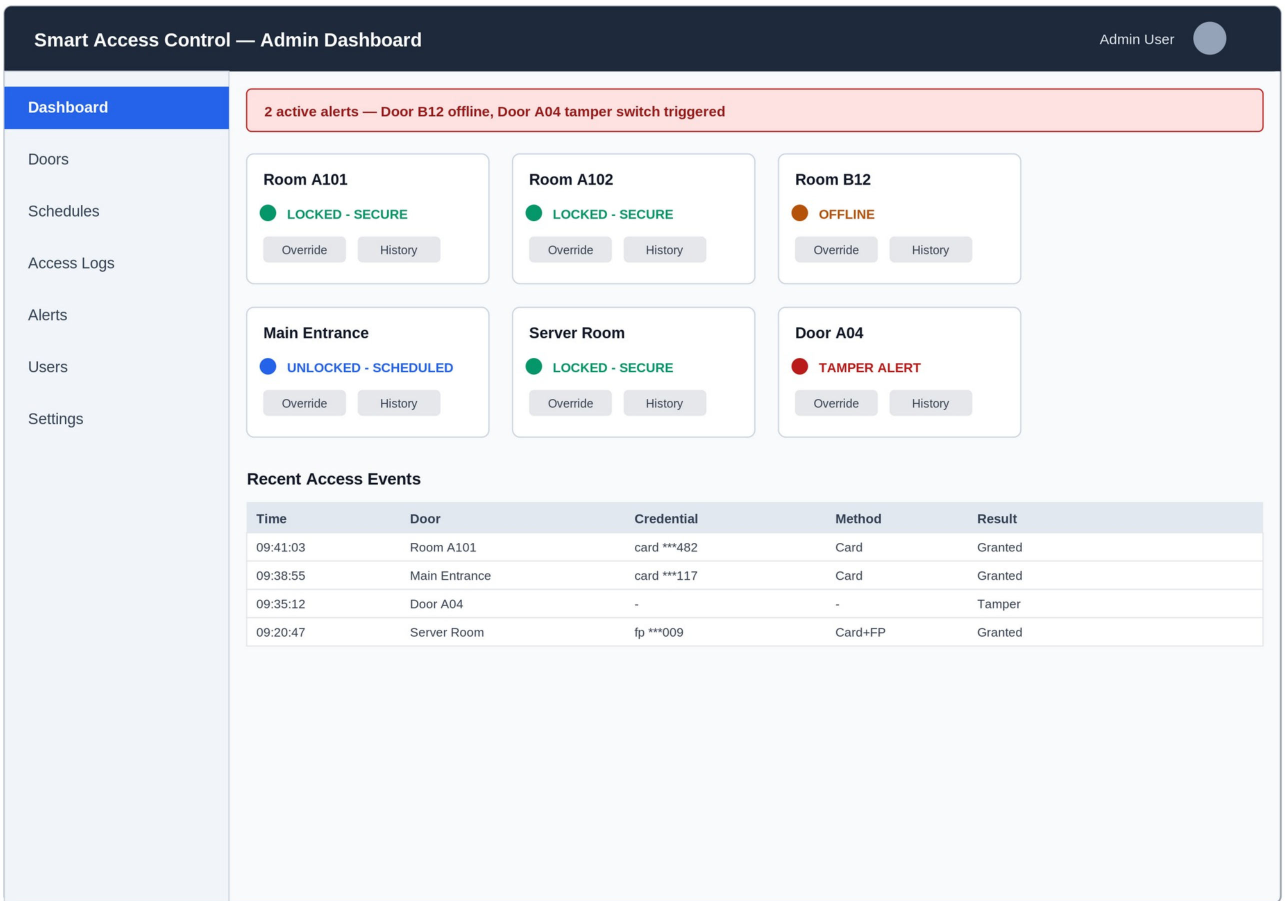

Figure 5 — Admin dashboard wireframe (Dashboard view).

8. Open Questions for Design Review

- Confirm PoE vs. RS-485-only for the backbone — affects gateway hardware and per-node cost.
- Confirm which rooms require the biometric second factor vs. card-only (drives fingerprint module BOM quantity).
- Confirm registrar timetable data format/access — determines whether Phase 3's scheduling import is a real integration or a structured mock.
- Confirm campus IT's stance on new devices on the network (VLAN, firewall rules for MQTT broker and gateway).
- Confirm fire-safety-mandated egress doors with campus facilities before finalizing which doors are fail-safe vs. fail-secure.